

Tuza's conjecture for line and total graphs

Samuel Massé

8 October 2026

Abstract

For a finite simple graph G , let $\nu_\Delta(G)$ be the maximum number of pairwise edge-disjoint triangles and let $\tau_\Delta(G)$ be the minimum number of edges meeting every triangle. We prove $\tau_\Delta(G) \leq 2\nu_\Delta(G)$ when G is the line graph or the total graph of any finite simple graph. The proofs use local triangle packings in incidence cliques and edge colorings obtained from Euler tours. The factor two is sharp over line graphs. For every cubic triangle-free graph F , we also prove the exact identity $\tau_\Delta(\text{Tot}(F)) = \nu_\Delta(\text{Tot}(F)) = 5|V(F)|/2$. The results are formalized in Lean 4.

1 Introduction

A *triangle packing* in a graph G is a collection of pairwise edge-disjoint triangles, and a *triangle transversal* is a set of edges meeting every triangle. Their maximum and minimum cardinalities are denoted by $\nu_\Delta(G)$ and $\tau_\Delta(G)$, respectively. Tuza conjectured that

$$\tau_\Delta(G) \leq 2\nu_\Delta(G) \tag{1}$$

for every finite graph G [7].

Every transversal must use a different edge to meet each triangle of an edge-disjoint packing, so $\nu_\Delta(G) \leq \tau_\Delta(G)$. Conversely, the union of the edges of a maximum packing is a transversal: any triangle it missed could be added to the packing. This gives the elementary bound $\tau_\Delta(G) \leq 3\nu_\Delta(G)$. Tuza's conjecture asks whether the factor three can always be reduced to two. The complete graph K_4 already shows that two would be best possible (Figure 1).

We establish (1) for two graph operators built from the incidence structure of a root graph. The *line graph* $L(H)$ has vertex set $E(H)$, with two vertices adjacent when the corresponding root edges meet. The *total graph* $\text{Tot}(F)$ has vertex set $V(F) \dot{\cup} E(F)$: it contains F on the original vertices and $L(F)$ on the edge-vertices, with each original vertex also adjacent to its incident edge-vertices. Figure 2 illustrates both constructions on a path.

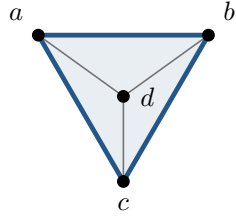
Theorem 1.1. *If H is a finite simple graph, then*

$$\tau_\Delta(L(H)) \leq 2\nu_\Delta(L(H)).$$

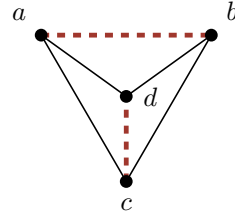
The factor two cannot be improved over the class of line graphs.

Theorem 1.2. *If F is a finite simple graph, then*

$$\tau_\Delta(\text{Tot}(F)) \leq 2\nu_\Delta(\text{Tot}(F)).$$

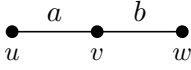


A packing of size one

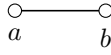


A transversal of size two

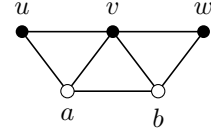
Figure 1: Packing and covering in K_4 . The highlighted triangle abc is a maximum packing, since any two triangles share an edge. Deleting the two dashed edges ab and cd destroys every triangle; deleting any one edge leaves a triangle. Hence $(\nu_\Delta, \tau_\Delta) = (1, 2)$.



Root graph H



Line graph $L(H)$



Total graph $\text{Tot}(H)$

Figure 2: The two graph operators on the path $u - v - w$, with edges $a = uv$ and $b = vw$. Solid dots represent original vertices; open circles represent root edges. The total graph has the three triangles uav , vab , and vbw : the middle one lies among the incidences at v , while the other two each use an original edge.

For cubic triangle-free roots, we obtain the following exact refinement.

Corollary 1.3. *If F is a finite simple cubic triangle-free graph, then*

$$\tau_\Delta(\text{Tot}(F)) = \nu_\Delta(\text{Tot}(F)) = \frac{5}{2}|V(F)|.$$

The main difficulty is to coordinate local choices. The root edges incident with a vertex become a clique in the line graph. These incidence cliques may share vertices, but their edge sets are disjoint, so triangle packings chosen inside them can be combined. Covers need more care: a root triangle gives a line-graph triangle with one edge in each of three different cliques, which independent local covers may miss.

A single red-blue coloring of the root edges supplies this coordination. We cover pairs of incident edges of the same color. Among any three root edges, two have the same color, so this selection hits both kinds of line-graph triangle. The coloring must also be locally balanced to keep the cover small; alternating colors along Euler tours achieves this with one controlled exception. In total graphs the original vertices provide the freedom to restore local balance, while bridge triangles supply an additional packing. A Hall assignment gives a sharp cover when the root is cubic and triangle-free.

The line-graph inequality for triangle-free roots appears in [4, Corollary 3.3.3]; Theorem 1.1 removes that hypothesis. Nearby results establish Tuza's inequality under a maximum average degree bound [6]; a recent preprint treats maximum degree at most seven [1]. Other results improve the coefficient for certain K_4 -free graphs [5]. Those hypotheses do not cover the unbounded-degree classes considered here. We are not aware of a previous statement of either theorem in the full scope given above.

The alternating Euler-tour coloring method is classical. Guruswami used it to study maximum cuts in line and total graphs [2]; Hanaka, Kobayashi and Sone give a later account of the dummy-vertex construction [3, Section 6]. Our use of this method pairs

the resulting covers with triangle packings. The odd Eulerian seam requires the explicit repair in Lemma 3.2; the total-graph argument instead balances each enlarged incidence clique by its original vertex.

All graphs in the paper are finite and simple; empty and disconnected roots are allowed. Simplicity ensures that two distinct root edges have at most one common endpoint. It suffices to construct a transversal C and a packing $\mathcal{P}$ in the same graph satisfying

$$\tau_\Delta(G) \leq |C| \leq 2|\mathcal{P}| \leq 2\nu_\Delta(G).$$

Neither constructed object needs to be optimal. The complete-graph bound uses a three-part construction and strong induction, with explicit packings for orders below 21; no exact formula for complete-graph packing numbers is needed.

2 Common tools

2.1 Clique packings

Splitting a clique into two nearly equal parts and deleting edges within the parts leaves a bipartite graph, hence destroys every triangle. We compare the cost $t(d)$ of this cover with the maximum packing size $D(d)$: for $d \geq 0$, put

$$t(d) = \binom{d}{2} - \left\lfloor \frac{d^2}{4} \right\rfloor, \quad D(d) = \nu_\Delta(K_d).$$

Explicitly,

$$t(2r) = r(r-1), \quad t(2r+1) = r^2. \quad (2)$$

We use the following elementary lower bounds on $D(d)$. The basic complete-graph inequality is classical. Our finite bases use the cyclic construction attributed to Sebő in [4, Theorem 3.3.2], with one extra triangle at even orders at least six. A Latin-square recurrence extends these finite bases to all orders and supplies the even-order slack needed in the line-graph proof.

Lemma 2.1 (Local clique slack). *For every integer $d \geq 0$,*

$$t(d) \leq 2D(d). \quad (3)$$

If $d \geq 6$ is even, then

$$t(d) + 2 \leq 2D(d). \quad (4)$$

Proof. The construction has two levels: pack triangles between three equal parts, then add packings inside the parts. Since these use different edges, their sizes add. A finite collection of small packings starts the induction.

For $q > 0$, partition $3q$ vertices into three copies A, B, C of $\mathbb{Z}/q\mathbb{Z}$. Take the q^2 triangles

$$\{a_A, b_B, (a+b)_C\}, \quad a, b \in \mathbb{Z}/q\mathbb{Z}.$$

Any two coordinates determine the third, so distinct triangles share at most one vertex. They use only edges between parts. Adding a maximum packing inside each of A, B, C therefore gives

$$D(3q) \geq q^2 + 3D(q). \quad (5)$$

Also, $D(a) \leq D(b)$ for $a \leq b$, by embedding K_a into K_b .

For completeness, the following describes precisely the finite witnesses used for $0 \leq n < 21$, on vertices $0, \dots, n-1$. Use the empty list for $n \leq 2$. For $n \geq 3$, put $r = \lfloor n/2 \rfloor$ and begin with

$$\{\{i, j, r + ((i + j) \bmod r)\} : 0 \leq i < j < r\}.$$

If $n = 2r \geq 6$, add $\{r, r+1, r+2\}$. If $n = 2r+1$, add

$$\{2r, i, r + ((2i) \bmod r)\} \quad (i \in I_r), \quad I_r = \begin{cases} \{0, \dots, r-1\}, & r \text{ odd}, \\ \{0, \dots, r/2-1\}, & r \text{ even}. \end{cases}$$

Each listed set has three distinct vertices. For the initial triangles, sharing a vertex in each of the two halves forces the remaining index to agree. The extra even-order triangle uses only edges in the second half. For odd orders, multiplication by two is injective on I_r modulo r ; moreover, an initial triangle containing i cannot also contain $r + ((2i) \bmod r)$, since its other first-half index would then equal i . Thus every pair of distinct listed triangles shares at most one vertex. Their sizes p_n are

n	0	1	2	3	4	5	6	7	8	9	10
p_n	0	0	0	1	1	2	4	6	7	8	11

and

n	11	12	13	14	15	16	17	18	19	20
p_n	15	16	18	22	28	29	32	37	45	46

Direct substitution verifies $t(n) \leq 2p_n$ and, for even $n \geq 6$, $t(n) + 2 \leq 2p_n$.

Proceed by strong induction. For $n \geq 21$, write $n = 3q + s$ with $q \geq 7$ and $0 \leq s < 3$. The function t is nondecreasing, and

$$t(3q + s) + 2 \leq 2q^2 + 3t(q). \quad (6)$$

Indeed it suffices to take $s = 2$. When $q = 2r$, the right side minus the left side is $2r^2 - 6r - 2 \geq 0$ for $r \geq 4$; when $q = 2r+1$, it is $2r^2 - 4r - 4 \geq 0$ for $r \geq 3$. The induction hypothesis, (5), and monotonicity now give

$$t(n) + 2 \leq 2q^2 + 3t(q) \leq 2(q^2 + 3D(q)) \leq 2D(3q) \leq 2D(n).$$

This proves both assertions, including the required even-order slack. $\square$

2.2 Balanced edge colorings

Both cover constructions use a red-blue coloring of the root edges with nearly equal color degrees at each vertex. Alternating colors along an Euler tour pairs an entering edge with a differently colored leaving edge at each visit; the only possible mismatch is where the tour closes. The following lemma also handles components without an Euler tour by first adding a dummy vertex.

Lemma 2.2 (Alternating Euler coloring). *Let K be a connected graph.*

- (a) *If K has an odd-degree vertex, its edges can be colored red and blue so that the two color degrees are equal at every even-degree vertex and differ by one at every odd-degree vertex.*
- (b) *If every degree of K is even and $|E(K)|$ is even, the two color degrees can be made equal at every vertex.*
- (c) *If every degree of K is even and $|E(K)|$ is odd, the two color degrees can be made equal except at one chosen seam vertex, where they differ by two.*

Proof. For (a), add a new dummy vertex adjacent to every odd-degree vertex. The augmented graph is connected and Eulerian. Start an Euler tour at the dummy vertex and color successive tour edges alternately. At every original vertex the tour edges pair into differently colored entering and leaving edges. Any odd cyclic seam occurs at the dummy vertex. After deleting the dummy edges, the stated balances remain.

For (b) and (c), an isolated vertex needs no coloring. Otherwise alternately color an Euler tour of K . If the tour length is even, every visit pairs two different colors. If it is odd, the first and last edges have the same color and the unique discrepancy of two occurs at the starting vertex. $\square$

3 Line graphs

An edge of a line graph is a *wedge* in the root: two root edges meeting at a vertex. This common endpoint assigns each line-graph edge to a unique incidence clique.

Let H be a finite simple graph. For $v \in V(H)$, let W_v be the set of edges of $L(H)$ corresponding to unordered pairs of distinct root edges incident with v . Two adjacent edges of a simple graph have a unique common endpoint, and hence

$$E(L(H)) = \dot{\bigcup}_{v \in V(H)} W_v. \quad (7)$$

We call W_v the *wedge packet* at v .

Lemma 3.1 (Line-graph triangle classification). *Every triangle of $L(H)$ is exactly one of the following:*

- (i) *a star triangle formed by three root edges incident with one vertex v , whose three edges all belong to W_v ;*
- (ii) *a root triangle formed by the three edges of a triangle xyz in H , with one edge in each of W_x, W_y, W_z .*

Proof. Three distinct pairwise intersecting two-element sets either have a common element or are $\{x, y\}, \{x, z\}, \{y, z\}$. Applied to three pairwise adjacent root edges, these are precisely the two stated cases. $\square$

Figure 3 shows the distinction and the effect of a shared coloring. A root triangle also gives the simplest reason that independent local covers do not suffice: if H is just a triangle, each incidence clique is a K_2 with no triangle to cover, yet $L(H)$ is a triangle. The local choices must therefore be coordinated.

The star triangles in W_v form the triangle system of $K_{d_H(v)}$. Choose $D(d_H(v))$ edge-disjoint star triangles at every root vertex. The packet partition (7) shows that their union is a packing of size

$$P_0 = \sum_{v \in V(H)} D(d_H(v)). \quad (8)$$

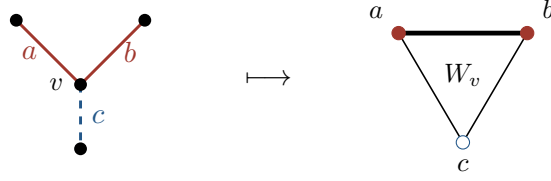
Write

$$T = \sum_{v \in V(H)} t(d_H(v)). \quad (9)$$

Lemma 2.1 gives $T \leq 2P_0$.

Apply Lemma 2.2 componentwise and select in W_v every wedge whose two root edges have the same color. Every star triangle is hit by the pigeonhole principle. Every root

(a) Three edges at one root vertex: a star triangle



(b) Three edges around a root triangle: three packets

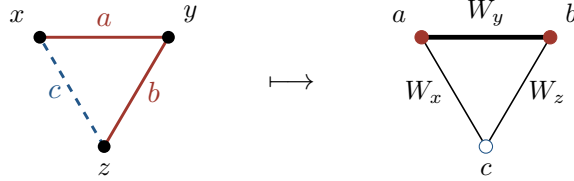


Figure 3: The two kinds of line-graph triangle. Solid root edges and filled line-graph vertices are red; dashed root edges and open vertices are blue. The bold line-graph edge joins two red vertices and is selected for the cover. In (a), all three line-graph edges lie in W_v ; in (b), they lie in three different packets. A single coloring covers both configurations because every triangle has two vertices of the same color.

triangle is hit because two of its three root edges have the same color. If the red and blue degrees at v are a and b , the local cost is

$$\binom{a}{2} + \binom{b}{2}.$$

By (2), this equals $t(d_H(v))$ when an even degree is balanced or an odd degree differs by one. At a difference-two even degree it equals $t(d_H(v)) + 1$. For example, at a degree-four vertex a two-red, two-blue split selects two wedges; a three-red, one-blue split selects three. The Euler exception therefore has a precise cost of one extra cover edge.

Consequently, every connected component has a cover of cost at most its contribution to T , except possibly a connected Eulerian component with an odd number of edges, where the cover may cost one extra edge. The following lemma absorbs this surcharge using existing packing slack, an independent local cover when the root is triangle-free, or one additional packed root triangle.

Lemma 3.2 (Odd Eulerian seam repair). *Let K be a connected Eulerian component of H with an odd number of edges. Put*

$$T_K = \sum_{v \in V(K)} t(d_H(v)), \quad P_K = \sum_{v \in V(K)} D(d_H(v)).$$

There is a triangle transversal C_K of $L(K)$ and a triangle packing $\mathcal{P}_K$ in $L(K)$ such that $|C_K| \leq 2|\mathcal{P}_K|$.

Proof. The alternating Euler cover has size at most $T_K + 1$. We distinguish three exhaustive cases.

If some vertex has degree at least six, its degree is even and (4) supplies two units of local packing slack. Together with (3) at every other vertex, this gives $T_K + 1 \leq 2P_K$.

Suppose next that K is triangle-free. Discard the global coloring. At each vertex, split its incident root edges into two equal parts and select the wedges lying inside a part. Since

there are no root triangles, these local selections cover every triangle and have total size exactly T_K . Thus $T_K \leq 2P_K$.

It remains that every positive degree is two or four and that K contains a root triangle xyz . Keep the alternating Euler cover of size at most $T_K + 1$. At a degree-two vertex the local packing is empty. At a degree-four vertex, $D(4) = 1$, and a triangle of the local K_4 can avoid any one prescribed wedge: if that wedge is ab , take the triangle on $\{a, c, d\}$. We may therefore choose the local maximum packings at x, y, z to avoid, independently, the three wedges of the triangle in $L(K)$ corresponding to xyz . Adding it to the local packing gives a packing of size $P_K + 1$. Hence

$$T_K + 1 \leq 2P_K + 1 \leq 2(P_K + 1).$$

These cases are exhaustive because, in the absence of a degree at least six, Eulerianity forces every positive degree to be two or four, after which the component is either triangle-free or contains a root triangle. $\square$

Proof of Theorem 1.1. Apply the alternating Euler cover to every component not requiring the seam repair, and Lemma 3.2 to every connected Eulerian odd-edge component. Line graphs and their triangle systems are disjoint across root components, so the component covers and packings may be united. This proves $\tau_\Delta(L(H)) \leq 2\nu_\Delta(L(H))$.

For sharpness, $L(K_{1,4}) \cong K_4$ has $(\nu_\Delta, \tau_\Delta) = (1, 2)$, as shown in Figure 1. $\square$

Remark 3.3. There is only one seam per connected Eulerian component with an odd number of edges. The triangle-free case can require a different cover: for a five-cycle, $T_K = P_K = 0$ and the line graph has no triangles, yet the alternating Euler cover selects one edge.

4 Total graphs

Let F be a finite simple graph. Write

$$d_v = d_F(v), \quad m = |E(F)|.$$

For every $v \in V(F)$, put

$$Q_v = \{v\} \cup \delta_F(v),$$

where $\delta_F(v)$ is the set of root edges incident with v , and an edge in this set is regarded as its edge-vertex in the total graph. The set Q_v induces K_{d_v+1} . Its edges are the incidence edges from v to its incident edge-vertices and the wedge edges between pairs of root edges meeting at v . These packet edge sets are pairwise disjoint. Original root edges lie outside all of them. For a root edge uv , let $B_{uv} = \{u, uv, v\}$ be its *bridge triangle*; here uv inside the vertex set denotes the edge-vertex.

The original vertex in Q_v restores local balance. An Euler coloring leaves a color-degree difference of at most two; adding v to the smaller side makes the difference at most one, even at a seam. For example, a three-red, one-blue split becomes three against two when v is blue. Selecting all original root edges costs a further m edges, but their bridges supply m triangles for the packing. In Figure 2, these bridges are uav and vbw .

Proof of Theorem 1.2. Apply Lemma 2.2 componentwise to color the root edges red and blue. At every odd-degree vertex the color degrees differ by one, and at every even-degree vertex they differ by at most two.

In each Q_v , retain these colors on the edge-vertices and color the original vertex v with the less frequent color, breaking ties arbitrarily. Adding one vertex to the minority side changes a discrepancy of zero, one, or two into one, zero, or one, respectively. Thus Q_v is balanced and its within-color edges number exactly $t(d_v + 1)$.

Let C consist of these within-color edges in every Q_v , together with all m original root edges. This gives a single red-blue coloring of all vertices of $\text{Tot}(F)$: each edge-vertex keeps its root-edge color, and each original vertex has its chosen minority color. Every monochromatic edge of $\text{Tot}(F)$ belongs to C . Indeed, an original-original edge was selected outright, an incidence edge belongs to its original vertex's packet, and a wedge belongs to the packet at the unique common root endpoint. Every triangle has a monochromatic edge by the pigeonhole principle, so C is a triangle transversal. Counting the packet contributions and original edges gives

$$|C| \leq m + \sum_v t(d_v + 1). \quad (10)$$

For a packing, take all m bridge triangles and, at each v , a packing of $D(d_v)$ triangles among the edge-vertices in $\delta_F(v)$. Distinct bridges use distinct original and incidence edges. The additional triangles use only wedge edges, whose packets are mutually disjoint. Thus their union is one packing $\mathcal{P}$ of size

$$|\mathcal{P}| = m + \sum_v D(d_v) \leq \nu_\Delta(\text{Tot}(F)). \quad (11)$$

The identity and degree-sum bound

$$t(d+1) - t(d) = \left\lfloor \frac{d}{2} \right\rfloor, \quad \sum_v \left\lfloor \frac{d_v}{2} \right\rfloor \leq m \quad (12)$$

follow from (2) and the degree sum. Therefore

$$\tau_\Delta(\text{Tot}(F)) \leq |C| \leq 2m + \sum_v t(d_v) \leq 2|\mathcal{P}| \leq 2\nu_\Delta(\text{Tot}(F)),$$

where the penultimate inequality is Lemma 2.1. $\square$

4.1 Exact values for cubic triangle-free roots

For a cubic triangle-free root, each packet is a K_4 , whose triangles are covered by two opposite edges (Figure 1). One of these is an incidence edge and also covers its bridge, so the corresponding original edge need not be selected. Distinct choices at the root vertices give one saving per vertex. Hall's theorem ensures that these choices can be made simultaneously.

Lemma 4.1 (Distinct incident-edge representatives). *If a finite simple graph F has minimum degree at least two, there is an injective map $p : V(F) \rightarrow E(F)$ such that $p(v)$ is incident with v for every vertex v .*

Proof. Apply Hall's theorem to the vertex-edge incidence bipartite graph. For $S \subseteq V(F)$, let $N(S)$ consist of the root edges incident with at least one vertex of S . Counting incidences gives

$$2|S| \leq \sum_{v \in S} d_F(v) \leq 2|N(S)|,$$

because a root edge has at most two endpoints in S . Hence $|S| \leq |N(S)|$ for every S , which is precisely Hall's condition. $\square$

Proof of Corollary 1.3. Put $n = |V(F)|$. Then $m = 3n/2$ and $D(3) = 1$, so the packing in (11) has $3n/2 + n = 5n/2$ triangles.

Choose p by Lemma 4.1. At each v , write its three incident edges as $p(v), a_v, b_v$. In the packet $Q_v \cong K_4$, select the incidence edge $vp(v)$ and the wedge $a_v b_v$. These two edges form a perfect matching of Q_v , so every triangle in the packet meets one of them. Also select every original edge outside $p(V(F))$.

For a bridge B_e , if $e = p(v)$ for some v , its selected incidence edge $vp(v)$ meets the bridge; otherwise its original edge is selected. Since F is triangle-free, every total triangle is a packet triangle or a bridge. To see this directly, three original vertices would give a root triangle; two original vertices force their joining edge-vertex; one original vertex forces the other two edge-vertices into its packet; and three edge-vertices are classified by Lemma 3.1, whose root-triangle case is excluded. The selection is therefore a transversal. Its size is at most

$$2n + (m - |p(V(F))|) = 2n + (m - n) = m + n = 5n/2,$$

using the injectivity of p . Finally, every transversal contains a distinct edge for each member of an edge-disjoint packing, so

$$\frac{5n}{2} \leq \nu_{\Delta}(\text{Tot}(F)) \leq \tau_{\Delta}(\text{Tot}(F)) \leq \frac{5n}{2}. \quad \square$$

5 Concluding remarks

The factor two is sharp for line graphs. We do not determine the best factor for total graphs, although the cubic triangle-free case has equality between packing and transversal numbers. Extensions to multigraph roots, generalized line graphs, or quasi-line graphs require additional arguments and are not claimed here.

Formal verification. Theorems 1.1 and 1.2, including line-graph sharpness, and Corollary 1.3 are formalized in Lean 4 with Mathlib. The formalization implements the clique packings, Euler colorings, componentwise seam repair and incident-edge assignment used above. The source code and instructions for checking the proofs are available at

<https://github.com/samuelfmasse/tuza-line-total-graphs>.

The accompanying proof version is commit `33a75de`, which pins Lean 4.35.0-rc4 and its Mathlib dependency. The repository gives the correspondence between the paper’s results and the formal declarations. Their proofs use propositional extensionality, classical choice and quotient soundness, with no additional axioms.

AI assistance. The author acknowledges the assistance of OpenAI’s GPT-6 Astra in developing the proofs, checking arguments, searching the literature, preparing the exposition, and constructing the Lean formalization.

References

- [1] Anish Gupta. Tuza’s conjecture for graphs of maximum degree at most seven, 2026. Preprint, arXiv:2608.06538v1.
- [2] Venkatesan Guruswami. Maximum cut on line and total graphs. *Discrete Applied Mathematics*, 92(2–3):217–221, 1999. doi:10.1016/S0166-218X(99)00056-6.
- [3] Tesshu Hanaka, Yasuaki Kobayashi, and Taiga Sone. A (probably) optimal algorithm for bisection on bounded-treewidth graphs. *Theoretical Computer Science*, 873:38–46, 2021. doi:10.1016/j.tcs.2021.04.023; see also Section 6 of arXiv:2002.12706v2.
- [4] Andrea Munaro. *Sur quelques invariants classiques et nouveaux des hypergraphes*. PhD thesis, Université Grenoble Alpes, 2016.
<https://andreamunaro.github.io/munaro-thesis.pdf>.
- [5] Andrea Munaro. Triangle packings and transversals of some K_4 -free graphs. *Graphs and Combinatorics*, 34(4):647–668, 2018. doi:10.1007/s00373-018-1903-y.
- [6] Gregory J. Puleo. Tuza’s conjecture for graphs with maximum average degree less than 7. *European Journal of Combinatorics*, 49:134–152, 2015.
doi:10.1016/j.ejc.2015.03.006.
- [7] Zsolt Tuza. A conjecture on triangles of graphs. *Graphs and Combinatorics*, 6(4):373–380, 1990. doi:10.1007/BF01787705.